

Statement of Purpose

Amazon Machine Learning Summer School

As a Computer Science undergraduate at IIIT Jabalpur, my technical journey has been driven by a fascination with how robust engineering and intelligent data processing converge to solve complex problems. I began by building a strong foundation in distributed systems and software engineering, actively contributing to open-source protocols like Braidpool. This experience taught me the rigor required to design scalable architectures, but my curiosity naturally gravitated toward how we can extract meaningful intelligence from massive, unstructured data environments.

To pursue this, I focused heavily on data science fundamentals, studying statistical analysis, probability distributions, and regression models. This foundational knowledge culminated in *Genesis_KB*, an automated Retrieval-Augmented Generation (RAG) pipeline I architected. The project's goal was to ingest dense, unstructured technical content from decentralized network conferences and transform it into a structured, highly searchable knowledge base. Building *Genesis_KB* required bridging traditional data ingestion pipelines with modern LLM capabilities, teaching me firsthand the practical challenges of vector search, contextual retrieval, and applied AI application design.

While I have successfully built and integrated applied AI systems, critical gaps remain in my knowledge. I currently operate largely as an AI integrator; I want to evolve into an AI architect. I lack a deep, under-the-hood understanding of training, fine-tuning, and mathematically optimizing foundational machine learning models from scratch. Furthermore, while I understand distributed systems, I lack enterprise-level visibility into how machine learning models are deployed, monitored, and scaled in highly demanding production environments. The Amazon ML Summer School, with its focus on industry-applied ML and infrastructure, is the exact bridge I need to cross this gap.

I believe my unique technical background makes me an ideal fit for this program. I am not approaching machine learning purely from a theoretical standpoint; I bring the engineering rigor of an open-source contributor who knows how to write clean, production-ready code in C++, Python, and Rust. I possess a self-driven track record of identifying complex problems

and building end-to-end systems to solve them. I am eager to absorb Amazon's customer-obsessed approach to machine learning and apply these learnings to build resilient, intelligent systems at scale.